

1. What are "Things" in IoT?

Definition

In the Internet of Things (IoT), Things are physical objects that can sense, process, communicate, and interact with their environment through the Internet.

These devices contain:

Sensors, Actuators, Microcontroller, Memory, Communication module, Power supply

Important Points

- Sensors collect information.
- Actuators perform actions.
- Things are connected through the Internet.
- They communicate with other devices automatically.
- They make systems smart and automated.

Example

Smart watch, Smart bulb, Smart parking sensor, Smart irrigation system. Things in IoT are smart physical devices that use sensors and actuators to collect data, communicate, and perform actions over the Internet.

2. Classification Methods of Sensors

Classification Method	Types / Description
External Energy	Active (requires external energy) vs. Passive (self-powered)
Placement	Contact vs. Non-contact
Contact (Distance)	Physical contact required vs. no contact needed
Application	Industrial, Medical, Environmental, Automotive, etc.
Measuring Scale	Analog vs. Digital output
Sensing Mechanism	Electrical, Optical, Magnetic, Chemical, Thermal
Sensing Parameter	Temperature, Pressure, Light, Motion, Humidity, etc.

3. Different Sensors Based on Sensing Parameter

Sensing Parameter	Example Sensors
Temperature	Thermistor, Thermocouple, RTD, LM35
Pressure	Piezoelectric, Barometric, Strain Gauge
Light / Optical	LDR, Photodiode, Phototransistor, IR Sensor
Motion / Proximity	PIR Sensor, Ultrasonic, Capacitive, Inductive
Humidity	Capacitive Humidity Sensor, Resistive Humidity Sensor
Gas / Chemical	MQ Series, Electrochemical Sensor
Sound / Acoustic	Microphone, Piezoelectric Sensor
Acceleration / Gyro	Accelerometer (MEMS), Gyroscope
Magnetic	Hall Effect Sensor, Magnetometer
Position / Displacement	Encoder, LVDT, GPS

Easy Mnemonic for the 7 Classification Methods

"Every Person Can Apply Many Smart Skills"

E → External Energy | P → Placement | C → Contact (Distance)

A → Application | M → Measuring Scale | S → Sensing Mechanism | S → Sensing Parameter

This format is compact, exam-friendly, and much easier to memorize than long notes.

4. Actuator

An Actuator is an output device that receives a control signal and converts it into a physical action such as movement, force, heat, pressure, or rotation.

Easy to Remember

Sensor → Senses Actuator → Acts

Examples

Motor, Solenoid, Hydraulic cylinder, Valve

5. Classification Methods of Actuators

Classification Method	Types / Description
Motion	Linear vs. Rotary motion
Power Output	Low power vs. High power
Output Type	Mechanical, Thermal, Hydraulic, Pneumatic, Electrical
Area of Application	Industrial, Medical, Automotive, Robotics
Energy Type	Electric, Hydraulic, Pneumatic, Thermal, Magnetic

6. Different Actuators Based on Energy Type

Energy Type	Examples	Advantages
Electric	DC Motor, Servo Motor, Stepper Motor, Solenoid	Easy to control, Clean, Precise
Hydraulic	Hydraulic Cylinder, Hydraulic Motor	High force, Heavy-duty applications
Pneumatic	Pneumatic Cylinder, Air Motor	Fast, Clean, Simple
Thermal / Shape Memory	SMA Actuator, Bimetallic Strip	Compact, Silent
Magnetic	Magnetostrictive Actuator, Voice Coil	High precision, Fast response

7. What is a Smart Object in IoT?

A Smart Object is a physical device with embedded technology that can sense, process, communicate, and perform actions automatically. It is the basic building block of IoT.

Components of a Smart Object

1. Sensor(s) and/or Actuator(s)

- Sensors collect data.
- Actuators perform actions.

2. Processing Unit

- Usually a Microcontroller (MCU).

Functions:

- Collects sensor data
- Processes data
- Controls actuators
- Controls communication

3. Memory

Stores:

- Program code
- User data
- Sensor readings
- Usually uses Flash Memory.

4. Communication Unit

Allows communication with:

- Other smart objects
- Internet
- Cloud

Uses: Wi-Fi, Bluetooth, ZigBee, LoRa

5. Power Source

Provides power to all components.

Examples: Battery, Solar cell, Power adapter

Characteristics of Smart Objects

Has sensors and actuators | Processes information | Stores data | Communicates through networks | Powered by a power source

Present Trends in Smart Objects

- Size is becoming smaller.
- Power consumption is decreasing.
- Processing power is increasing.
- Communication capability is improving.
- Communication standards are becoming more common.

Easy Memory Tricks

Sensor = Sense (Input) | Actuator = Action (Output)

Smart Object = Sensor + Actuator + Processor + Memory + Communication + Power

Thing in IoT = Smart Object connected to the Internet

8. IoT Network Access Technologies

IoT Network Access Technologies are the communication technologies that allow IoT devices (smart objects) to connect, exchange data, and communicate with each other or with the Internet.

Can be: Wired communication or Wireless communication

Today, wireless technologies are more popular because they are easy to install and allow devices to move freely.

Examples: WiFi, Bluetooth, ZigBee, LoRa, NB-IoT, Ethernet

9. Different Attributes (Communication Criteria) of Access Technologies

Communication criteria describe the characteristics used to compare different IoT communication technologies. There are six main communication criteria.

1. Communication Criteria

Communication Criteria are the properties or characteristics used to evaluate and compare different IoT communication technologies. These criteria help us choose the best communication technology for a specific IoT application.

2. Communication Range

Communication Range is the maximum distance between two devices where communication can occur successfully. Communication range is divided into three types.

Range Type	Distance	Technology Example
Short Range	< 10 meters	Bluetooth
Medium Range	Up to 100 meters	WiFi
Long Range	Kilometres	Cellular & LoRa

3. Frequency Band

A Frequency Band is the range of radio frequencies used for wireless communication. The radio spectrum is controlled by organizations like: ITU, FCC, TRAI

Frequency bands are divided into two types: Licensed Band and Unlicensed Band (ISM Band)

ISM means: Industrial, Scientific and Medical

These frequency bands are freely available for many wireless devices.

Common ISM Bands: 2.4 GHz, 5 GHz, 868 MHz, 915 MHz

India also allows: 865–867 MHz

4. Power Consumption

Power Consumption is the amount of electrical energy required for an IoT device to operate. Power consumption is important because most IoT devices run on batteries.

Power Supply Type	Description
Battery Powered	Most IoT devices; requires low power operation
Wired / AC Powered	Industrial devices; high power allowed

Why Low Power is Important?

Battery-powered IoT devices should: Consume less energy | Increase battery life | Reduce maintenance cost

Examples: ZigBee, Bluetooth Low Energy (BLE), LoRa, NB-IoT

Comparison of Power Consumption

Technology	Power Consumption
WiFi	Highest Power
Bluetooth	Medium
ZigBee	Low
LoRa / NB-IoT	Lowest Power

Easy Remember: WiFi → Highest Power | LoRa → Lowest Power

5. Topology

Topology is the physical or logical arrangement of devices in a communication network. It defines how devices connect and communicate. There are three main topologies.

Topology	Description	Example Layout
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Star	All devices connect to a central hub/ router	Device—Router—Device
Mesh	Devices connect to multiple neighbors; self-healing	A—B—C—D interconnected
Peer-to-Peer	Devices communicate directly without central node	A — B — C (direct links)

6. Constrained Devices

Constrained Devices are IoT devices with limited Memory, Processing power, Storage, and Battery. Because of these limitations, they cannot run complex software.

RFC 7228 divides constrained devices into three classes:

Class	RAM	Flash	Description
Class 0	< 10 KB	< 100 KB	Very Small — cannot run Internet protocols directly
Class 1	~ 10 KB	~ 100 KB	Medium — limited protocol support
Class 2	~ 50 KB	~ 250 KB	Powerful — can support most protocols

Easy Remember: Class 0 → Very Small | Class 1 → Medium | Class 2 → Powerful

7. Constrained Networks (LLNs)

A Constrained Network is a network made up of constrained devices with limited Energy, Memory, Processing, and Bandwidth. These are also called LLNs (Low-Power and Lossy Networks).

Lossy means data packets may sometimes be lost because of weak wireless links.

Characteristics of Constrained Networks

Characteristic	Description
Low Bandwidth	Very limited data transfer rate
High Packet Loss	Weak links cause frequent packet loss
Low Power	Devices run on batteries with minimal energy
Small Packet Size	IEEE 802.15.4 MTU = 127 Bytes (vs IPv6 min 1280 Bytes)

Important Note

IPv6 Minimum MTU = 1280 Bytes | IEEE 802.15.4 MTU = 127 Bytes

Therefore, large IPv6 packets must be divided into smaller packets before transmission. This process is called Fragmentation.

10. Comparison of Common IoT Access Technologies

Technology	Range	Frequency	Power	Topology	Data Rate
WiFi	~100m	2.4/5 GHz	High	Star	High
Bluetooth	~10m	2.4 GHz	Medium	P2P/Star	Medium
BLE	~10m	2.4 GHz	Very Low	Star	Low
ZigBee	~100m	2.4 GHz	Low	Mesh/Star	Low
LoRa	~15 km	Sub-GHz	Very Low	Star	Very Low
NB-IoT	Wide Area	Licensed	Very Low	Star	Very Low

Ethernet	~100m	Wired	High	Star	Very High
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11. Security in IoT

IoT Security is the process of protecting IoT devices, networks, and data from unauthorized access, attacks, and misuse. Its main goal is to ensure that data is: Confidential | Accurate | Available | Safe from attackers

Why is IoT Security Important?

IoT contains millions of smart devices connected through the Internet. These devices usually have Low processing power, Small memory, Limited battery, and Low bandwidth. Therefore, traditional Internet security methods (TCP/IP security) are not suitable for IoT.

Security Requirements in IoT

Requirement	Description
Confidentiality	Only authorized parties can read the data
Integrity	Data has not been modified during transmission
Availability	System and data are accessible when needed
Authentication	Verify the identity of devices/users
Authorization	Ensure only allowed access to resources
Non-repudiation	Sender cannot deny sending the message

Common Security Attacks in IoT

Attack Type	Description
Eavesdropping	Attacker listens to communication without authorization
Man-in-the-Middle	Attacker intercepts and possibly alters communication
Replay Attack	Attacker captures and resends valid messages
DoS / DDoS	Overloads device/network to deny service
Sybil Attack	Attacker creates multiple fake identities
Physical Attack	Attacker physically tampers with the device

12. Cryptography

Cryptography is the technique of converting readable data (plaintext) into unreadable data (ciphertext) to protect information from attackers. It provides secure communication over insecure networks.

Types of Cryptography

Type	Keys Used	Advantages	Examples
Symmetric Key	Same key for encryption & decryption	Faster, Less computation, Suitable for IoT	DES, Triple DES, AES
Asymmetric Key	Public Key (shared) + Private Key (secret)	More secure, Supports digital signatures	RSA, Diffie-Hellman, ECC

Confidentiality

Confidentiality means that only the sender and the intended receiver can read the transmitted

message. It protects data from unauthorized users. Achieved by Symmetric Encryption and Asymmetric Encryption.

Message Integrity

Message Integrity ensures that the message has not been modified during transmission. If even one bit changes, integrity is lost.

How is Integrity Checked?

Using a Hash Function.

Hash Function

A Hash Function converts a message of any size into a fixed-length value, called a Hash Value or Message Digest. Even a small change in the message produces a completely different hash value.

Properties of Hash Function

Property	Description
One-way	Cannot reverse the hash to get original message
Deterministic	Same input always gives same hash output
Collision-resistant	Two different inputs should not produce same hash
Fixed output size	Output length is always fixed regardless of input size

Popular Hash Functions: MD5, SHA-1

Message Authentication

Message Authentication verifies: (1) The sender is genuine. (2) The message has not been modified.

Message Authentication Code (MAC)

A MAC is created by combining Message + Secret Authentication Key. Only the sender and receiver know the secret key. Therefore, MAC provides Authentication and Integrity.

Note: MAC means Message Authentication Code or Message Integrity Code (MIC). Do not confuse it with MAC Layer = Medium Access Control Layer

Digital Signature

A Digital Signature is an electronic signature created using Asymmetric Key Cryptography. It proves Sender authenticity, Message integrity, and Non-repudiation.

Uses: Software verification, Online banking, Digital certificates, Electronic documents

Public Key Infrastructure (PKI)

Public Key Infrastructure (PKI) is a framework that verifies whether a public key really belongs to the claimed person or organization. It makes public keys trustworthy.

Components

Component	Role
Certificate Authority (CA)	Issues and signs digital certificates
Registration Authority (RA)	Verifies identity before CA issues certificate
Digital Certificate	Binds public key to an entity identity
Certificate Revocation List (CRL)	Lists certificates that are no longer valid

End-Point Authentication

End-point authentication is the process of verifying the identity of a communicating device or user.

Authentication Method	Description
Password-based	Simple username/password verification

Token-based	Physical or digital token used for authentication
Certificate-based	Digital certificate verifies identity
Nonce-based	Random number used once to prevent replay attacks

A Nonce is a random number used only once during authentication. Purpose: Prevent replay attacks | Ensure every authentication session is unique.

13. IoT PHY Layer (Physical Layer)

The Physical Layer (PHY) is responsible for transmitting and receiving data over the wireless channel. IEEE 802.15.4 PHY manages: Radio communication | Channel selection | Signal transmission | Energy management

Functions of PHY Layer

Function	Description
Channel Selection	Selects the appropriate radio channel for communication
Signal Transmission	Converts digital data into radio signals for transmission
Signal Reception	Receives radio signals and converts them back to digital data
Energy Detection	Measures signal energy on the channel
Link Quality Indication	Indicates the quality of the received signal
Clear Channel Assessment (CCA)	Checks if the channel is free before transmitting

Reliability Techniques

To reduce interference, IEEE 802.15.4 uses: DSSS (Direct Sequence Spread Spectrum), UWB (Ultra-Wideband), CSS (Chirp Spread Spectrum). These techniques improve Signal-to-Noise Ratio (SNR) and Communication reliability.

14. IoT MAC Layer

The MAC (Medium Access Control) Layer controls how devices access the wireless communication channel.

Functions of MAC Layer

Function	Description
Channel Access	Controls when each device can transmit (CSMA/CA)
Frame Generation	Creates and manages MAC frames
Synchronization	Provides beacon-based time synchronization
Association / Disassociation	Manages device joining and leaving the network
Security	Provides encryption and authentication at the MAC level

Supported Topologies: Star, Peer-to-Peer, Cluster Network

Device Types

Device Type	Description
Full Function Device (FFD)	Can act as coordinator; supports all MAC functions

Reduced Function Device (RFD)	Simple device; can only communicate with FFD
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Types of MAC Frames: Data Frame, ACK Frame, Beacon Frame, Command Frame

Security in IoT PHY & MAC Layer

IEEE 802.15.4 provides several security suites at the MAC layer.

Security Suite	Protection Provided
AES-CTR	Confidentiality only (Encryption)
AES-CBC-MAC	Authentication + Integrity only
AES-CCM	Confidentiality + Authentication + Integrity (All three)

Easy Trick: CTR → Encryption → Confidentiality

CBC-MAC → Authentication + Integrity

CCM = CTR + CBC-MAC → All three (Confidentiality, Authentication, Integrity)

Data Frame with Security Header

When security is enabled, IEEE 802.15.4 adds an Auxiliary Security Header. It contains:

Field	Purpose
Security Control	Indicates security level and key mode
Frame Counter	Prevents replay attacks; increments with each frame
Key Identifier	Identifies which key is used for this frame

15. Security-related Information in IoT

Security-related information is the additional information attached to a message that helps protect it from attacks such as replay attacks, unauthorized access, and message modification. Different security suites (AES-CTR, AES-CBC-MAC, AES-CCM) require different security information to provide different levels of protection.

Protection Against Replay Attack

What is a Replay Attack?

A Replay Attack is an attack in which an attacker captures a valid message and sends (replays) it again later to trick the receiver.

Example

Suppose Alice sends: "Open the smart door." An attacker records this message and sends it again after one hour. The smart door opens again, even though Alice did not send the command again. This is called a Replay Attack.

How IEEE 802.15.4 Prevents Replay Attack

Field	Role in Preventing Replay Attack
Frame Counter	Unique incrementing number; receiver rejects old counters
Nonce / IV	Unique value per frame; ensures different ciphertext each time

What is Frame Counter?

A Frame Counter is a unique number added to every transmitted message. Each new message gets a larger Frame Counter value. Receiver accepts only messages with a higher Frame Counter.

If an attacker replays Message 1 (Frame Counter = 100), Receiver already has 102. Therefore,

Message 1 is rejected.

Security Suites

Suite	Security Provided
AES-CTR	Confidentiality only
AES-CBC-MAC-32	Integrity + Authentication (4-byte tag)
AES-CBC-MAC-64	Integrity + Authentication (8-byte tag)
AES-CBC-MAC-128	Integrity + Authentication (16-byte tag)
AES-CCM-32	Confidentiality + Integrity + Authentication (4-byte tag)
AES-CCM-64	Confidentiality + Integrity + Authentication (8-byte tag)
AES-CCM-128	Confidentiality + Integrity + Authentication (16-byte tag)

Easy Trick: CTR → Confidentiality | CBC-MAC → Integrity + Authentication | CCM = CTR + CBC-MAC → Provides all three protections.

Nonce / Initialization Vector (IV)

A Nonce (Number Used Once) or Initialization Vector (IV) is a unique value used during encryption. It ensures that the same plaintext always produces different ciphertexts.

Why is Nonce Used? It provides: Replay protection | Semantic security | Different ciphertext for identical messages

Types of IV

IV Type	Description
Random IV	Generated randomly for each message
Counter-based IV	Uses incrementing counter (Frame Counter + Source Address)

Block Counter

If a message is very large, it is divided into 16-byte blocks. Each block has its own Block Counter. This helps encrypt every block differently.

16. Access Control

Access Control is the process of allowing only authorized users or devices to access the IoT network. Unauthorized devices are denied access.

How Access Control Works

IEEE 802.15.4 MAC Layer uses Access Control List (ACL).

What is ACL?

ACL means Access Control List. It is a table stored inside the radio chip. Maximum entries: 255. Each entry usually represents one destination address.

Information Stored in ACL

Field	Description
Device Address	IEEE address of the trusted device
Security Key	Encryption key used with this device
IV / Replay Counter	Initialization vector and replay protection counter

Example

Suppose a Smart Home has: Phone, Smart Lock, Smart Camera. ACL stores these three device addresses. If another unknown device tries to connect, ACL rejects it.

17. Added Security with TSCH

TSCH means Time Slotted Channel Hopping. It is an enhancement of IEEE 802.15.4e that improves Security, Reliability, and Energy efficiency.

TSCH divides communication into time slots. Each device transmits only during its assigned time slot. It also changes the communication channel periodically.

Advantages of TSCH

Less interference | Lower packet loss | Better synchronization | Better security

TSCH Security Improvements

IEEE 802.15.4e introduces:

Feature	Description
ASN (Absolute Slot Number)	Every communication slot gets a unique number; used as Frame Counter
Time-dependent security	Security tied to time slot; provides replay protection and semantic security
Reserved Bits	Two reserved bits used for additional security control

What is ASN?

ASN means Absolute Slot Number. Every communication slot gets a unique number. ASN is used as the Frame Counter.

Benefits of ASN: Time-dependent security | Replay protection | Semantic security

Reserved Bits

Two reserved bits are used in the security control field for additional security configuration.

18. Adaptation Layer Protocol (6LoWPAN)

6LoWPAN stands for IPv6 over Low-Power Wireless Personal Area Networks. It allows IPv6 communication over low-power wireless IoT devices.

Why is 6LoWPAN Needed?

IPv6 packets are large. IEEE 802.15.4 packets are very small. Therefore, 6LoWPAN compresses IPv6 packets.

Functions of 6LoWPAN

Function	Description
Header Compression	Compresses large IPv6 headers to fit in small IEEE 802.15.4 frames
Fragmentation & Reassembly	Splits large IPv6 packets into small fragments; reassembles at destination
Neighbor Discovery	Adapted version for low-power networks
Mesh Addressing	Supports multi-hop routing in mesh networks

Supported Header Types

Header Type	Description
LOWPAN_HC1	First generation header compression

LOWPAN_IPHC	Improved header compression (replaces HC1)
LOWPAN_NHC	Next header compression for UDP, extension headers
LOWPAN_MESH	Mesh addressing header for multi-hop routing
LOWPAN_FRAG	Fragmentation header for large packets

Example

Suppose IPv6 packet size = 1280 Bytes and IEEE 802.15.4 packet = 127 Bytes. 6LoWPAN divides the packet into many small packets.

19. Security in 6LoWPAN

6LoWPAN itself does not define its own security mechanism. Instead, it relies on IEEE 802.15.4 MAC Layer Security and Network Layer Security.

Important RFCs

RFC	Topic
RFC 4861	Neighbor Discovery for IPv6
RFC 3971	SEND — Secure Neighbor Discovery
RFC 3972	CGA — Cryptographically Generated Addresses

What is SEND?

SEND means Secure Neighbor Discovery. It protects Neighbor Discovery messages from attackers.

What is CGA?

CGA means Cryptographically Generated Address. IPv6 address is generated using cryptographic methods. This prevents address spoofing.

20. RPL (Routing Protocol for Low-Power and Lossy Networks)

RPL is the routing protocol specially designed for Low-Power and Lossy Networks (LLNs) used in IoT. It finds the best route between IoT devices.

RPL Control Messages

Message	Full Name	Purpose
DIO	DODAG Information Object	Sends routing information downward
DIS	DODAG Information Solicitation	Requests routing information
DAO	Destination Advertisement Object	Advertises destination routes upward
DAO-ACK	DAO Acknowledgement	Confirms receipt of DAO
CC	Consistency Check	Checks security and consistency

What is DODAG?

DODAG means Destination-Oriented Directed Acyclic Graph. It is a tree-like routing structure. Each DODAG has one Root and a Unique ID.

Routing Parameters

RPL uses: Link Cost | Node Rank | Node Status. These are evaluated using Objective Function (OF).

21. Security in RPL

RPL supports three security modes.

Security Mode	Description
Unsecured Mode	No security; no encryption or authentication
Pre-installed Mode	Security keys are pre-installed before deployment
Authenticated Mode	Join Authentication Server verifies new devices

Security Fields in RPL

Field	Description
Security Level	Defines the type of protection (encryption, authentication)
Algorithm	Specifies cryptographic algorithm (e.g., AES-CCM)
Key Identifier	Identifies which key is used
Counter	Replay protection counter

Security Services Provided by RPL

Confidentiality | Integrity | Authentication | Replay Protection

22. Easy Memory Tricks

Topic	Memory Trick
Security Suites (CTR)	CTR → Confidentiality
Security Suites (CBC-MAC)	CBC-MAC → Integrity + Authentication
Security Suites (CCM)	CCM → Confidentiality + Integrity + Authentication
Replay Protection	Frame Counter + Nonce + Counter = Replay Attack Protection
Access Control	ACL = Address + Key + IV + Replay Counter
6LoWPAN Functions	Compress → Discover → Auto-configure
RPL Messages	DIO → Sends DIS → Requests DAO → Advertises DAO-ACK → Confirms CC → Checks

23. CoAP (Constrained Application Protocol)

CoAP (Constrained Application Protocol) is a lightweight application layer protocol specially designed for low-power, low-memory, and constrained IoT devices. It enables communication between IoT devices using the request-response model, similar to HTTP, but with much lower overhead.

Why is CoAP Used?

IoT devices have Small memory, Low processing power, Low battery, and Low bandwidth. HTTP is too heavy for these devices. Therefore, CoAP is used because it is lightweight and efficient.

Features of CoAP

Feature	Description
Lightweight	Small header size; minimal overhead
UDP-based	Uses UDP instead of TCP for lower overhead

RESTful	Supports GET, POST, PUT, DELETE like HTTP
Asynchronous	Supports asynchronous request-response
Multicast Support	Can send messages to multiple devices at once
Observe Option	Client can subscribe to resource updates from server
Low Power	Designed for battery-powered IoT devices

CoAP Request-Response Model

A client sends a request to a server. The request can be:

Method	Purpose
GET	Retrieve resource from server
POST	Create new resource on server
PUT	Update existing resource on server
DELETE	Delete resource from server

Reliability in CoAP

CoAP uses the Stop-and-Wait mechanism.

Stop-and-Wait Mechanism

- Sender sends one packet.
- Sender waits for ACK.
- If ACK is not received, the sender retransmits the packet.
- This ensures reliable communication.

Exponential Backoff

If ACK is not received, the sender waits longer before each retransmission. Example: 1 second → 2 seconds → 4 seconds → 8 seconds. This technique is called Exponential Backoff. It reduces network congestion.

Types of CoAP Responses

Response Type	Description
Piggy-backed Response	Server replies immediately with both ACK and response data in one message
Separate Response	Server first sends ACK, then sends response data in a separate message later

Example

Smartphone requests temperature from a sensor. Sensor immediately replies with the temperature. → Piggy-backed Response. If processing takes time, Sensor first sends ACK, then later sends the temperature. → Separate Response

24. CoAP Security

CoAP Security protects CoAP messages from unauthorized access, modification, and attacks. CoAP uses DTLS (Datagram Transport Layer Security) to provide security over UDP.

What is DTLS?

DTLS (Datagram Transport Layer Security) is the UDP version of TLS. It provides: Confidentiality | Authentication | Integrity for CoAP messages.

Security Modes of CoAP

Mode	Description	Key Type
NoSec	No encryption, no authentication, no integrity. Used only in testing.	None
PreSharedKey (PSK)	Both devices share the same secret key. Uses Symmetric Cryptography. Provides Authentication, Encryption, Integrity.	Symmetric
Raw Public Key (RPK)	Each device has Public Key + Private Key. Authentication based on public keys. No certificates required.	Asymmetric
Certificates	Each device has Public Key + Private Key + X.509 Certificate. Certificates issued by trusted CA. Highest level of trust.	Asymmetric + PKI

ECC in CoAP

CoAP uses ECC (Elliptic Curve Cryptography) because IoT devices have limited resources. ECC provides the same security as RSA using much smaller keys.

ECC Supports

ECC Feature	Description
ECDSA	Elliptic Curve Digital Signature Algorithm — used for digital signatures
ECDHE	Elliptic Curve Diffie-Hellman Ephemeral — used for key exchange

PreSharedKey Cipher Suite

CoAP uses TLS_PSK_WITH_AES_128_CCM_8. It provides: Symmetric encryption | Authentication | 8-byte Nonce | 8-byte Integrity Code

25. Security Requirements at Operational Levels

Security requirements are divided into three operational levels.

Level	Definition	Key Requirements
A. Information Level	Protects information during storage and transmission.	Confidentiality, Integrity, Availability, Privacy
B. Access Level	Controls who can use the IoT system.	Authentication, Authorization, Access Control, Identity Management
C. Functional Level	Ensures the IoT system continues working even during failures or attacks.	Reliability, Resilience, Fault Tolerance, Self-healing

26. Security Mechanisms for IoT

Security mechanisms are techniques used to satisfy IoT security requirements.

A. Encryption

Encryption converts readable data (Plaintext) into unreadable data (Ciphertext). It provides Confidentiality.

Type	Description	Examples
Symmetric Encryption	Same key for encryption and decryption; fast; suitable for IoT	AES, DES, Triple DES
Asymmetric Encryption	Public/Private key pair; more secure; higher computation	RSA, ECC, Diffie-Hellman

B. Authentication & Integrity

Authentication verifies identity. Integrity ensures the message has not changed.

Technique	Description
MAC (Message Authentication Code)	Combines message + secret key; provides authentication and integrity
Digital Signature	Asymmetric key based; provides authentication, integrity, non-repudiation
Hash Functions	Produces fixed-length message digest; verifies integrity

Hash Algorithms: MD5, SHA-160, SHA-256, SHA-512

C. Lightweight Cryptography

Lightweight cryptography consists of cryptographic algorithms specially designed for IoT devices with limited memory, processing power, and battery.

Lightweight Block Ciphers

Cipher	Block Size	Key Size	Note
PRESENT	64-bit	80/128-bit	Very compact; suitable for Class 0 devices
SIMON	Variable	Variable	NSA designed; hardware efficient
SPECK	Variable	Variable	NSA designed; software efficient
CLEFIA	128-bit	128/192/256-bit	Sony; compact structure
HIGHT	64-bit	128-bit	Designed for low-resource environments

Lightweight Hash Functions

Hash Function	Output Size	Note
PHOTON	Variable	Very lightweight; designed for IoT tags
SPONGENT	Variable	Sponge-based; hardware efficient
QUARK	Variable	Ultra-lightweight; suitable for RFID

27. Secure Hardware

Secure Hardware protects IoT devices against physical attacks and side-channel attacks.

Side Channel Attack

Instead of breaking encryption, an attacker observes Power consumption, Timing, or Electromagnetic signals to discover secret keys.

PUF (Physically Unclonable Function)

A PUF generates a unique fingerprint for every IoT device based on tiny manufacturing differences in

the chip. No two chips produce exactly the same fingerprint.

	Detail
Advantages	Unique device identity Difficult to clone Strong authentication
Disadvantage	Higher power consumption

28. Intrusion Detection System (IDS)

An Intrusion Detection System (IDS) monitors IoT devices and networks to detect suspicious or malicious activities.

Why Lightweight IDS?

Traditional antivirus software is too heavy for IoT devices. Therefore, IoT uses Lightweight IDS.

Lightweight IDS Monitors

Parameter Monitored	What It Detects
CPU Usage	Sudden 100% usage without valid task may indicate malware
Memory Usage	Unexpected memory spikes indicate potential attack
Network Traffic	Unusual packet frequency or volume indicates intrusion
Power Consumption	Abnormal power use can signal unauthorized activity

Example

If CPU usage suddenly becomes 100% without any valid task, IDS detects possible malware.

29. Security in Commercial IoT Devices

General Architecture

Most IoT devices contain Two Microcontrollers:

Microcontroller	Role
Application MCU	Runs the main application code and sensors
Security / Communication MCU	Handles security operations, encryption, and communication

ARM Cortex-M Series

Most commercial IoT devices use ARM Cortex-M Microcontrollers.

Type	Features
Cortex-M0 / M0+	Ultra-low power; very small; Class 0/1 devices
Cortex-M3	Low power; good performance; Class 1/2 devices
Cortex-M4	DSP extensions; FPU; medium performance
Cortex-M7	High performance; hardware FPU; Class 2 devices
Cortex-M33	TrustZone security; hardware security extensions

Security Limitation

Most Cortex-M processors do not include: Hardware Random Number Generator | Hardware AES Engine. Therefore, Cryptography is implemented in software or using a dedicated security co-processor.

Examples

Device / Chip	Security Features
STM32 (STMicroelectronics)	Some include hardware AES and RNG
nRF52 (Nordic Semiconductor)	Hardware AES-CCM accelerator included
ATECC608A (Microchip)	Dedicated security co-processor; hardware ECC; key storage
ESP32	Hardware AES accelerator; SHA accelerator; RNG

30. Quick Exam Summary

Topic	Key Points
CoAP	Lightweight application protocol Uses UDP RESTful architecture Request–Response model CON and NON messages Stop-and-Wait reliability
CoAP Security	DTLS NoSec PreSharedKey RawPublicKey Certificates ECC (ECDSA, ECDHE)
Operational Security Levels	Information Level Access Level Functional Level
Security Mechanisms	Encryption Authentication Integrity Lightweight Cryptography Secure Hardware (PUF) IDS
Commercial IoT Security	Two microcontrollers ARM Cortex-M family Hardware AES support in some devices

Easy Memory Tricks — Final Summary

Trick	Meaning
CoAP = HTTP for IoT	Lightweight + UDP based protocol
DTLS = TLS for UDP	Security for CoAP over UDP
PSK = Same Secret Key	PreSharedKey mode uses symmetric key
RPK = Public Key without Certificate	RawPublicKey mode uses key pair only
Certificate = Public Key + X.509 Certificate	Highest trust mode with CA-issued certificate
ECC = Smaller Keys, Same Security	Elliptic Curve Cryptography for resource-constrained IoT
PUF = Unique Fingerprint of a Chip	Hardware-level device authentication
IDS = Watches CPU, Memory, and Network for Attacks	Lightweight intrusion detection for IoT